

Implementing HDBSCAN:

Insights, Limitations, and Applications on Real-World Datasets

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Flat vs Hierarchical vs Centroid vs Density

	Flat	Hierarchical
Centroid	KMeans	Agglomerative Clustering
Density	DBSCAN	?

Flat vs Hierarchical vs Centroid vs Density

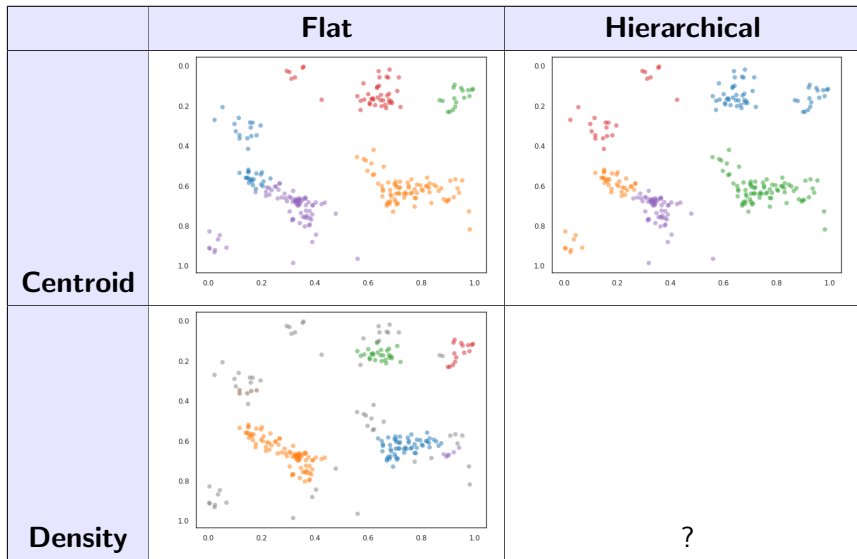


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Density Estimation

1. Core Distance:

- Distance to the k -th nearest neighbor.
- Identifies dense and sparse regions.

2. Mutual Reachability Distance:

$$d_{\text{mreach-}k}(a, b) = \max\{\text{core}_k(a), \text{core}_k(b), d(a, b)\}$$

- Prevents over-connecting sparser regions.

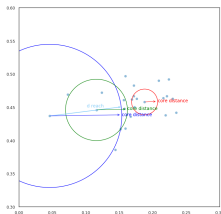


Figure: Example of mutual reachability distance.

Minimum Spanning Tree (MST)

What is the MST?

- A tree that connects all data points (vertices) with the minimum possible total edge weight.
- Captures the essential structure of the dataset.

Construction:

- Vertices: Data points.
- Edge weights: Mutual reachability distances.
- Algorithm: Prim's or Kruskal's algorithm.

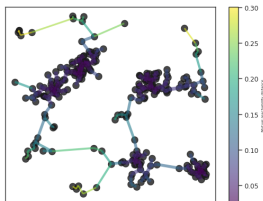


Figure: Example of a Minimum Spanning Tree.

Constructing the Cluster Hierarchy

Cluster Hierarchy:

- Sort edges by increasing weight.
- Merge clusters as thresholds decrease.

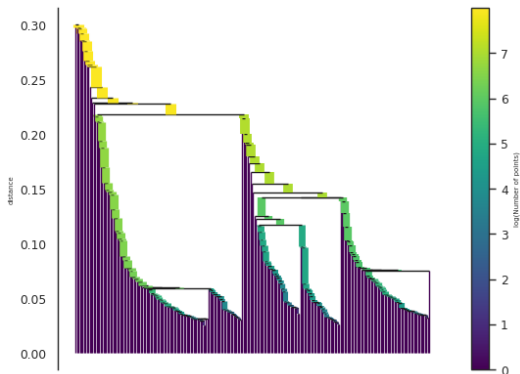


Figure: Dendrogram showing cluster hierarchy.

Condensing the Cluster Hierarchy

Condensation Process:

- Introduce a minimum cluster size parameter.
- Eliminate insignificant clusters.
- Track points that leave clusters.

Visualization:

- Condensed tree with cluster sizes.
- Lines become thinner as points leave clusters.

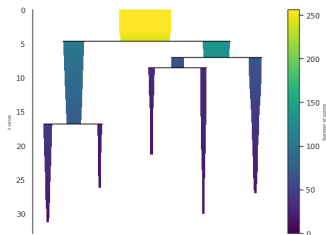


Figure: Example of a condensed tree.

Extracting Meaningful Clusters

Cluster Stability:

$$\text{Stability} = \sum_{p \in \text{cluster}} (\lambda_p - \lambda_{\text{birth}})$$

- Measures cluster persistence.

Selection and Labeling:

- Points in selected clusters are assigned unique labels.
- Points outside clusters are labeled as noise (-1).

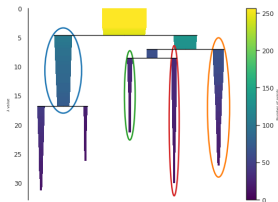
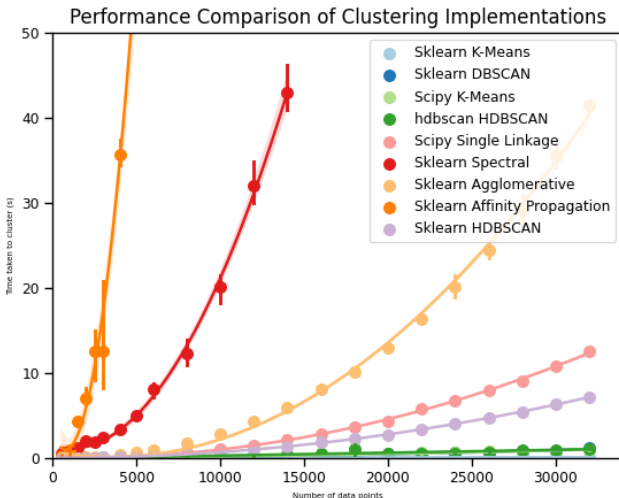


Figure: Selection of stable clusters.

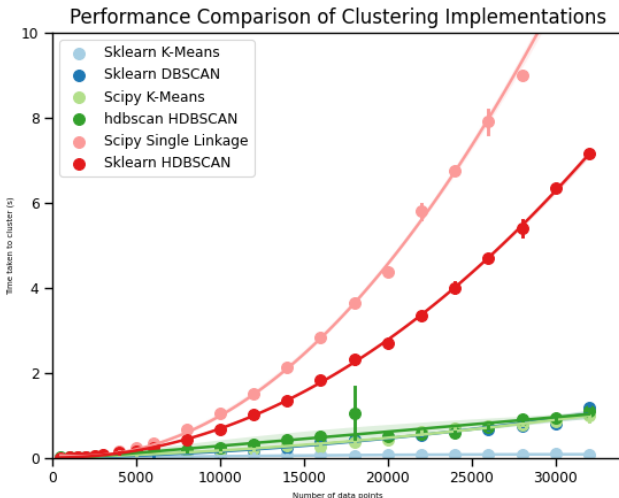
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Performance Scaling



Performance Scaling



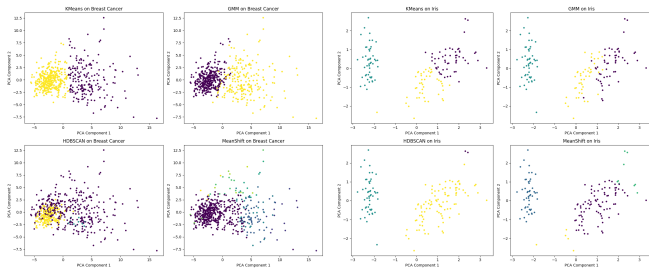
Performance Scaling



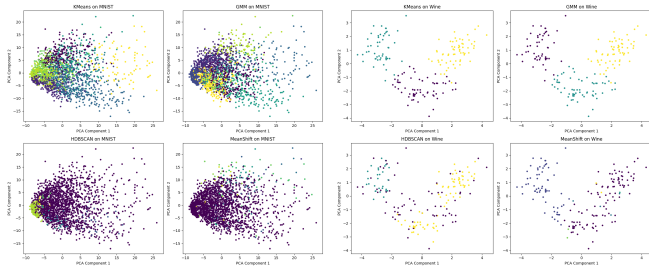
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Limitations



Noise Sensitivity (Breast Cancer Data) Spherical Clusters (Iris Data)



High Dimensional Data (MNIST) Uniform Density (Wine Data)

Clustering Performance Comparison

Dataset	Method	ARI	Silhouette
Breast Cancer	GMM	0.774016	0.314489
	KMeans	0.653625	0.343382
	MeanShift	0.269276	0.168140
	HDBSCAN	0.118554	-0.047853
Iris	GMM	0.903874	0.374165
	KMeans	0.645147	0.456535
	HDBSCAN	0.563751	0.494863
	MeanShift	0.536264	0.353130
MNIST	GMM	0.282834	0.012530
	KMeans	0.249380	0.017750
	HDBSCAN	0.045981	-0.170825
	MeanShift	0.000724	0.137828
Wine	KMeans	0.863599	0.282837
	GMM	0.863599	0.282837
	MeanShift	0.376718	0.200543
	HDBSCAN	0.265957	0.098521

Table Notes: This table presents clustering performance using ARI and Silhouette scores across datasets and methods.

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HDBSCAN: A Robust Clustering Tool

HDBSCAN was designed to handle large-scale data efficiently, excelling in detecting complex clusters and noise. Efficient to uncover structures in noisy, and unstructured datasets

Strengths:

- Adapts to varying densities and handles noise well.
- Delivers high-quality results with minimal parameter tuning
- Excels with real-world datasets (e.g., 32,000 Bluesky tweets)

Limitations:

- Sensitive to high-dimensional data (curse of dimensionality)
- Not always the fastest or simplest algorithm.
- Requires careful parameter tuning for optimal performance

BlueSky

- Downloaded 2M tweets and down-sample to 32k to reduce exploration time

Approach

- Encode tweets using a Language Model
- UMAP to reduce the dimension of the embeddings
- HDBSCAN + TF-IDF to find both clusters and topics

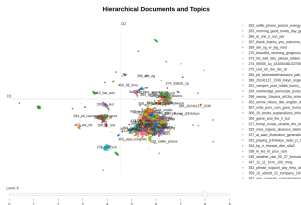


Figure: Topics indentified by HDBSCAN

Gaussian Mixture Model (GMM) for Clustering

Overview:

- Probabilistic clustering algorithm based on Gaussian distributions
- Represents clusters as Gaussian components defined by:
 - **Mean (μ)**: Center of the cluster.
 - **Covariance Matrix (Σ)**: Shape and orientation of the cluster
 - **Mixing Coefficient (π)**: Proportion of points in the cluster

How it Works:

- ➊ **E-step**: Estimate the probability of each point belonging to each cluster.
- ➋ **M-step**: Update parameters (μ , Σ , π) to maximize the likelihood.
- ➌ Assign points to clusters based on maximum probability

Advantages:

- Models elliptical clusters with varying sizes and shapes
- Provides soft clustering

Limitations:

- Sensitive to initialization and overfitting.
- Assumes Gaussian distribution, which may not fit all datasets

Dimensionality Reduction: PCA and UMAP

Principal Component Analysis (PCA):

- A linear dimensionality reduction technique.
- Projects data onto directions of maximum variance.
- Captures global structure but may lose local relationships.
- **Use case:** Simplifies data for clustering or visualization.

Uniform Manifold Approximation and Projection (UMAP):

- A non-linear dimensionality reduction method.
- Preserves both local and global structures in data.
- More effective for high-dimensional, complex datasets.
- **Use case:** Visualizing clusters in datasets like MNIST.

Comparison:

- PCA is computationally faster but assumes linear relationships.
- UMAP is more versatile but computationally intensive.

Curse of Dimensionality

What is the Curse of Dimensionality?

- As the number of dimensions increases:
 - Data points become more sparse in the feature space.
 - Distances between points lose discriminatory power.
- Impacts density-based algorithms like HDBSCAN.

Effects:

- Challenges in clustering due to unreliable distance metrics.
- Higher computational cost.
- Reduced interpretability of models.

Solutions:

- Dimensionality reduction techniques like PCA and UMAP.
- Feature selection to retain only important dimensions.
- Regularization to mitigate overfitting in high dimensions.

Clustering Evaluation: ARI and Silhouette Score

Adjusted Rand Index (ARI):

- Measures the similarity between predicted and true cluster labels.
- Adjusted for chance to ensure a value close to 0 for random labels.
- **Range:** $[-1, 1]$
 - 1: Perfect agreement.
 - 0: Random labeling.
 - < 0 : Worse than random.

Silhouette Score:

- Measures how similar a data point is to its cluster compared to others.
- **Range:** $[-1, 1]$
 - 1: Well-clustered and far from other clusters.
 - 0: Overlapping clusters.
 - < 0 : Misclassified points.
- **Use case:** Evaluates compactness and separation of clusters.

Comparison:

- ARI evaluates alignment with ground truth.
- Silhouette assesses intrinsic cluster quality without labels.